

Skill- 7

```
himasree@LAPTOP-41COT5AA:~$ cd ~/OSLab
himasree@LAPTOP-41COT5AA:~/OSLab$ sudo apt update
[sudo] password for himasree:
Sorry, try again.
[sudo] password for himasree:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1009 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1275 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [214 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1208 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [294 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1689 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [243 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.2 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1449 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [338 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [336 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [11.3 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1580 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [359 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [45.7 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [13.1 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
```

```
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Co...
Fetched 11.3 MB in 21s (535 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
112 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 112 not upgraded.
himasree@LAPTOP-41COT5AA:~/OSLab$ nano process_path.c
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc process_path.c -o process_path
himasree@LAPTOP-41COT5AA:~/OSLab$ ls
Apipes          excel           fcfs            parser.c        program25.c     q5.c            signal5.c
Apipes.c        excel.c         fcfs.c          permissions     program3.c      renamed.txt     signal6.c
Apipes2.c       excel.c.save   fifoRead        permissions.c   program31.c     sample.txt     signal7.c
Appipes2        exec           fifoRead.c      process_path    program31.c     scde.c         scdeuling
FIFO_CREATE     exp1.c         file1.txt        process_path.c  program31.c     server          skill12.q2
FIFO_CREATE.c   exp1.c         file_D           producer_consumer  program32.c     server.c       skill2.q2.c
FIFO_READ       exp2.c         form             producer_consumer.c  program32.c     server.c.save  skill3.1
FIFO_READ.c     exp2.c         form.c           program          program33.c     server1.c      skill3.1.c
FIFO_WRITE      exp3.c         histoiroy        program02.c     program4.c      server1.c      source.txt
FIFO_WRITE.c    exp3.c         histoiroy.c      program02.c     program41.c     shell          std.c
LinuxLab        exp4.c         ls_grep          program1         program41.c     shell.c        trace.txt
MYProject       exp4.c         ls_grep.c        program1.c      program41.c     signal1        wait
MyProject       exp5.c         ls_grep.c.save   program2         program42.c     signal1.c     waitpid.c
algorithms      exp5.c         myfifo           program2.3      program42.c     signal2.c     zombie
client          exp6.c         p1.c             program2.3.c    q2.c            signal2.c     zombie.c
client.c        exp6.c         p1.c             program2.c      q3.c            signal3.c
client1         exp7.c         p2.c             program23.c     q4.c            signal4.c
client1.c       exp7.c         p2.c             program24.c     q5.c            signal5
client_to_server exp8.c         p3.c             program25
copy.c          exp8.c         p3.c
copy.txt        exp9.c         p4.c
destination.txt exp9.c         parser
```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ ./process_path
===== PATH VARIABLE =====
PATH = /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.10.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/Program Files/nod ej:/mnt/c/Users/Himasree/AppData/Local/Programs/Python/Python311/Scripts:/mnt/c/Users/Himasree/AppData/Local/Programs/Python/Python311:/mnt/c/Users/Himasree/.local/bin:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/Himasree/AppData/Local/Microsoft/WindowsApps:/mnt/c/S10C/bin:/mnt/c/Users/Himasree/Desktop/college/Microsoft VS Code/bin:/mnt/c/Users/Himasree/AppData/Roaming/npm:/snap/bin

===== SEARCH DIRECTORIES =====
Directory: /usr/local/sbin
Directory: /usr/local/bin
Directory: /usr/sbin
Directory: /usr/bin
Directory: /sbin
Directory: /bin
Directory: /usr/games
Directory: /usr/local/games
Directory: /usr/lib/wsl/lib
Directory: /mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.10.0_x64__8wekyb3d8bbwe
Directory: /mnt/c/Program Files/Common Files/Oracle/Java/javapath
Directory: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
Directory: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
Directory: /mnt/c/Windows/system32
Directory: /mnt/c/Windows
Directory: /mnt/c/Windows/System32/Wbem

```

Enter command to locate: ls

===== COMMAND RESOLUTION =====

Command found: ls

Executable path: /usr/bin/ls

File exists: YES

Executable permission: YES

===== PROCESS SYNCHRONIZATION =====

Parent process created child.

Parent PID: 886

Child PID: 887

Parent is waiting for child using waitpid()...

Child process started.

Child PID: 887

Child is performing some work...

Child process completed.

Child exit status: 5

Enter command to locate: gcc

===== COMMAND RESOLUTION =====

Command found: gcc

Executable path: /usr/bin/gcc

File exists: YES

Executable permission: YES

===== PROCESS SYNCHRONIZATION =====

Parent process created child.

Parent PID: 890

Child PID: 891

Parent is waiting for child using waitpid()...

Child process started.

Child PID: 891

Child is performing some work...

Child process completed.

Child process has finished.

Child exited normally.

Child exit status: 5

himasree@LAPTOP-41COT5AA:~/OSLab\$ |

Enter command to locate: hello123

==== COMMAND RESOLUTION ====
Command 'hello123' not found in PATH.

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cat process_path.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <sys/stat.h>
#include <string.h>

#define MAX_PATH 4096

// Function to find executable in PATH
char *find_command(const char *command)
{
    char *path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return NULL;
    }

    // Make a copy because strtok modifies the string
    char *path_copy = malloc(strlen(path) + 1);

    if (path_copy == NULL)
    {
        perror("malloc");
        return NULL;
    }
}
```

```
    strcpy(path_copy, path);

    // Split PATH using ':'
    char *directory = strtok(path_copy, ":");

    while (directory != NULL)
    {
        char full_path[MAX_PATH];

        snprintf(full_path, sizeof(full_path),
                 "%s/%s", directory, command);

        // Check whether file exists and is executable
        if (access(full_path, X_OK) == 0)
        {
            char *result = malloc(strlen(full_path) + 1);

            if (result != NULL)
                strcpy(result, full_path);

            free(path_copy);
            return result;
        }

        directory = strtok(NULL, ":");
    }

    free(path_copy);
    return NULL;
}
```

```

}

int main()
{
    pid_t pid;
    int status;

    printf("==== PATH VARIABLE ====\n");

    // Retrieve PATH
    char *path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH = %s\n\n", path);

    // Parse PATH directories
    printf("==== SEARCH DIRECTORIES ====\n");

    char *path_copy = malloc(strlen(path) + 1);

    if (path_copy == NULL)
    {
        perror("malloc");
        return 1;
    }

```

```

    strcpy(path_copy, path);

    char *directory = strtok(path_copy, ":");

    while (directory != NULL)
    {
        printf("Directory: %s\n", directory);
        directory = strtok(NULL, ":");
    }

    free(path_copy);

    // Ask user for command
    char command[100];

    printf("\nEnter command to locate: ");
    scanf("%99s", command);

    // Locate command
    printf("\n==== COMMAND RESOLUTION ==== \n");

    char *location = find_command(command);

    if (location != NULL)
    {
        printf("Command found: %s\n", command);
        printf("Executable path: %s\n", location);

        // Verify permissions

```

```
// =====
// PROCESS SYNCHRONIZATION USING waitpid
// =====

printf("\n==== PROCESS SYNCHRONIZATION =====\n");

pid = fork();

if (pid < 0)
{
    perror("fork");
    return 1;
}

// Child process
if (pid == 0)
{
    printf("Child process started.\n");
    printf("Child PID: %d\n", getpid());

    printf("Child is performing some work...\n");

    sleep(2);

    printf("Child process completed.\n");

    exit(5);
}
```

```
// Parent process
else
{
    printf("Parent process created child.\n");
    printf("Parent PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);

    printf("Parent is waiting for child using waitpid()...\n");

    // Wait for specific child
    if (waitpid(pid, &status, 0) == -1)
    {
        perror("waitpid");
        return 1;
    }

    printf("Child process has finished.\n");

    // Check how child terminated
    if (WIFEXITED(status))
    {
        printf("Child exited normally.\n");
        printf("Child exit status: %d\n", WEXITSTATUS(status));
    }
    else if (WIFSIGNALED(status))
    {
        printf("Child terminated by signal.\n");
        printf("Signal number: %d\n", WTERMSIG(status));
    }
}
}
```

```
    }
}

return 0;
}
```